

Dundalk Institute of Technology

An Exploratory Analysis of Participation Patterns in European Citizens' Initiatives

Interim Report June 2026

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Contents

1	Introduction	4
1.1	Background	4
1.2	Research Problem	5
1.3	Research Questions	6
1.4	Project Relevance to Data Analytics	6
1.5	Technology and Research Process	7
1.6	Preliminary Hypotheses	8
1.7	Structure of the Report	8
2	Literature Review	9
2.1	The European Citizens' Initiative. Mobilization Strategies and Consequences	9
2.2	Petition Growth and Success Rates on the UK No. 10 Downing Street Website	10
2.3	Mobilizing Europe's citizens to take action on migration and climate change: behavioral evidence from 27 EU member states	10
2.4	Identifying patterns and recommendations of and for sustainable open data initiatives: a benchmarking-driven analysis of open government data initiatives among European countries	11
2.5	Direct democracy and equality: context is the key	12
2.6	Research Gap	13

3	Exploration of Data and Methods	14
3.1	Research Methodology	14
3.2	Data Collection	14
3.3	Ethical Considerations	16
3.4	Initial Exploratory Data Analysis	17
3.5	Data Preparation	17
3.6	Post Cleaning Exploratory Data Analysis	19
3.7	Initial Data Modelling	21
3.8	Initial Evaluation	22
3.9	Artefact Design	22
4	Proposed Future Analysis	23
	Bibliography	24
A	Appendices	26
A.1	Project Workflow Chart	26
A.2	Histogram of Signatures Per Capita	27
A.3	Histogram of GDP per Capita	28
A.4	Histogram of Unemployment Rate	29
A.5	Histogram of Population Density	30
A.6	Boxplot of GDP Per Capita	31

A.7 Countries with highest Signatures per Capita	32
A.8 Signatures per Capita vs Petition	33
A.9 Signatures per Capita vs GDP per Capita	33
A.10 Signatures per Capita vs Unemployment Rate	34
A.11 Signatures per Capita vs Tertiary Enrollment	35
A.12 Signatures per Capita vs Population Density	36
A.13 Pearson Heatmap	36
A.14 Spearman Heatmap	37
A.15 Kendall's Tau Heatmap	37
A.16 Github Repository	38
A.17 Mahara Portfolio	38

1 Introduction

1.1 Background

The European Citizens' Initiative (ECI) was introduced by the European Union in 2012 as a way for which citizens of the European Union can become more involved in democratic processes of the European Parliament and the development of policy and legislature (European Commission, n.d.). Each ECI is essentially a petition that allows citizens to propose or amend legislature or policies given that said ECI collects at least 1 million signatures and at least 7 member states reach the minimum threshold number of signatures. Each ECI is given a period of exactly 1 year to collect signatures. The ECIs that pass the required number of signatures while reaching the minimum threshold in 7 member states will be taken forward to the European Parliament. Since its introduction, many Initiatives have passed on a wide variety of issues including environmental protection, agriculture, healthcare, digital rights (European Commission, n.d.).

In the digital age, new opportunities have arisen that allow for greater data based analysis of civic participation. The Europa.eu website contains detailed publicly accessible data regarding the signature collection process of each Initiative including country level signatures and thresholds. This data provides a valuable resource for analysing patterns and participation rates across different member states and across different initiatives (Europa.eu, 2024).

While there exists research on the effectiveness of the ECI based on its impact on an institutional level or its overall democratic effectiveness (Weisskircher, 2019), there is still relatively little quantitative research based on participation behavior and country level patterns. For example, limited research has explored whether or not country level socio-economic factors influence participation rates within European Citizens' Initiatives.

This project aims to address this gap by collecting and analysing publicly available ECI signature data using web scraping techniques and then combining this data with

public datasets of external country level factors. This phase of the project focuses specifically on country level socio-economic factors such as GDP, Population, Unemployment rates etc. but aims to expand into other areas such as media literacy as well as examine possible petition level factors to get an overall bigger picture. The current resulting dataset allows for a comparative analysis of participation patterns across EU member states and compares them to the socio-economic factors that have been selected.

1.2 Research Problem

The success of a European Citizens' Initiative depends on how effective it can mobilise citizen support across multiple countries within the 1 year collection timeframe. However, rates of participation vary greatly between initiatives and countries. Some initiatives are able to surpass the 1 million signature mark and meet the minimum thresholds in at least 7 countries. While others fail to gain significant support. Furthermore, some countries are able to easily surpass their thresholds while some struggle to meet them. While having a population of 5.6 million, Finland achieved 552.56% of the threshold in the 2024 Stop Destroying Videogames Initiative, meanwhile Italy's population of 58.9 million only reached 143.77% of the threshold (*Stop Destroying Videogames*, 2024).

This threshold each country has is defined by it's number of Members of the European Parliament (Europa.eu, 2024). While it is for the most part aligned to a country's population, it still sees countries with the lowest populations struggling to reach the thresholds. Cyprus, Malta and Luxembourg successively fail to reach their thresholds in most Initiatives; 47.21%, 47.45% and 58.27% in the 2024 Stop Destroying Videogames Initiative respectively (*Stop Destroying Videogames*, 2024).

The factors that overall influence these differences in participation rates across countries remains unclear without a single root cause. While demographic and economic factors may play a role, petition specific factors such as topic relevance, campaign effectiveness, media attention and public interest may also play a significant part in ECI

outcomes.

This project overall seeks to investigate what influences participation patterns across different countries. This phase of the project for the interim report seeks to specifically investigate whether country level socio-economic factors can explain differences in participation rates in European Citizens' Initiatives.

1.3 Research Questions

This phase of the project is guided by the following research questions:

- **Research Question 1:** *How do participation levels differ across European Citizens' Initiatives?*
- **Research Question 2:** *To what extent do country level socio-economic factors explain variation in ECI participation?*
- **Research Question 3:** *Are differences in participation more strongly associated with petition specific characteristics than country level socio-economic factors?*

1.4 Project Relevance to Data Analytics

This project utilised various data analytics techniques to a real world democratic participation problem. This study incorporates web scraping, data cleaning and preparation, feature extraction, exploratory data analysis, statistical modelling and machine learning concepts into a data analytics workflow.

This project also illustrates how data from different sources can be incorporated and integrated to create a richer dataset that is capable of supporting deeper analysis. Combining web scraped ECI signature data with country level socio-economic variables. This

provides a practical example of data integration and feature engineering techniques common to many data analytics projects.

In addition, the project contributes to real world research regarding citizen participation behavior within the EU which may inform and contribute to the improvement of democratic processes in the EU. This research uses quantitative methods and publicly available data sources.

1.5 Technology and Research Process

This project follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology (Data Science PM, 2024). Web scraping was performed by using Python and the Playwright framework (Medium, 2025) to extract the tables containing country level signature data on the webpage of each European Citizens' Initiative. The scraped data of each Initiative was exported into a CSV file for initial Exploratory Data Analysis and then Data Preparation using Pandas.

An external country level statistics dataset was obtained from Kaggle. This dataset was also investigated in initial Exploratory Data Analysis and then cleaned and prepared and merged into the petition data to provide demographic and economic indicators for each country. Additional features were extracted by combining existing variables. The fully prepared data was then used for full Exploratory Data Analysis and Data Modelling using Python Libraries such as Pandas, NumPy, Matplotlib and Seaborn. Models were exported using Joblib (APXML, n.d.).

Preliminary statistical analyses included correlation analysis, linear regression models and Analysis of Variance (ANOVA) testing. These methods were used to investigate relationships between the participation metrics and the country level socio-economic variables.

1.6 Preliminary Hypotheses

Variance between Petitions

- H_0 : There are no significant differences in participation rates between petitions.
- H_1 : Participation rates differ significantly between petitions.

Raw Signature Counts

- H_0 : Population size and economic indicators do not explain a significant proportion of the variation in raw signature counts.
- H_1 : Population size and economic indicators explain a significant portion of the variation in raw signature counts.

Signatures Per Capita

- H_0 : Socio-economic indicators have no significant relationship with participation rates when participation is normalised by population.
- H_1 : Socio-economic indicators have limited explanatory power when participation is normalised by population.

1.7 Structure of the Report

This report is structured as follows. Chapter 1 covers the introduction and goals of the project. Chapter 2 covers a review of relevant literature in relation to European Citizens' Initiatives, online petition participation and the use of data analytics within democratic engagement research. Chapter 3 describes the data collection process, initial exploratory data analysis, data cleaning and preparation, post cleaning exploratory data analysis and

initial modelling results. Chapter 4 covers evaluation and planned future development of the project.

2 Literature Review

2.1 The European Citizens' Initiative. Mobilization Strategies and Consequences

Weisskircher (2019) investigates the role of social movements and mobilisation strategies within the European Citizens' Initiatives (ECIs). This study argues that successful Initiatives tend to focus their efforts within a small number of bigger EU member states rather than an EU wide mobilisation effort. Through conducting an in-depth analysis of one of the few initiatives that successfully surpassed 1 million signatures, the research shows that campaign organisation and mobilisation strategies are key determinants of success. This study also highlights that Initiatives are able to generate important political outcomes beyond just legislative change such as increased public awareness and institutional reform (Weisskircher, 2019).

These findings are relevant to this project as they suggest that participation levels may vary significantly between initiatives due to campaign specific factors rather than just demographic or socio-economic factors on its own. This supports the inclusion of multiple petitions within the dataset and provides justification for analysing differences between different initiatives using statistical techniques such as Analysis of Variance (ANOVA). Furthermore, this study also highlights that participation might be concentrated within the larger EU member states, this supports the investigation of population based variables for this dissertation (Weisskircher, 2019).

2.2 Petition Growth and Success Rates on the UK No. 10 Downing Street Website

In this paper, Hale, Margetts, and Yasseri (2013) analysed over 8,000 online petitions submitted through the UK Government’s No. 10 Downing Street petition platform. They used a large-scale data analytics approach to investigate petition growth patterns and identify key factors associated with petition success. Their findings revealed that successful petitions usually experience rapid growth during the early stages of the campaign and that signatures collected during the first day of the petition are strongly associated with overall campaign performance. This study also demonstrated that participation patterns are highly irregular and unequal, with a relatively small number of petitions obtaining substantial support while the majority of petitions receive limited engagement and fail to pass (Hale et al., 2013).

This research is highly relevant to the dissertation as it shows how petition data can be analysed using statistical methods to examine citizen participation. In a similar approach to this dissertation, the authors of the paper utilised large-scale behavioral data rather than survey responses, this provides direct evidence of public engagement. The study also highlights the importance of petition specific factors when explaining participation outcomes. The findings in this paper also support the inclusion of multiple initiatives in the dataset and the paper also provides a similar methodology as a foundation for the exploratory and predictive analyses undertaken in this dissertation (Hale et al., 2013).

2.3 Mobilizing Europe’s citizens to take action on migration and climate change: behavioral evidence from 27 EU member states

Giebler, Giesecke, Humphreys, Hutter, and Klüver (2025), investigate citizen participation across all 27 EU member States using behavioral data collected through the Eu-

ropean Parliament Election Study. The authors analysed support for petitions relating to migration and climate change and they examined whether country level factors could explain differences in participation between member states. This study found substantial differences in support levels across countries and policy issues. However the authors also reported that country level factors demonstrated limited explanatory power when attempting to explain participation behaviour (Giebler et al., 2025).

The findings by Giebler et al. (2025) are relevant to this dissertation as they align closely with the preliminary findings obtained during the exploratory data analysis and initial data modelling. The initial regression models developed in this dissertation also indicated weak relationships between participation rates and socio-economic factors such as GDP per Capita and unemployment rate. The study suggested that participation behaviour may be influenced more strongly by issue specific factors, political context and campaign factors rather than solely by country level factors on its own. These findings provide an important basis for interpreting weak regression performance observed in the initial modelling stage of this dissertation (Giebler et al., 2025).

2.4 Identifying patterns and recommendations of and for sustainable open data initiatives: a benchmarking-driven analysis of open government data initiatives among European countries

Lnenicka et al. (2023) examines the development and sustainability of open government data initiatives across European countries. Using a benchmarking driven analysis of open data frameworks and indices, the authors identified patterns that influence the effectiveness and long term sustainability of open government data systems. The study also highlights the importance of open data as a resource for research and development as well as transparency and evidence based decision making. It also highlights that public datasets can be integrated and analysed to generate new insights into social and

governmental processes (Lnenicka et al., 2023).

Although the study doesn't focus specifically on political participation or petitions, it nevertheless provided important justifications for the use of publicly available government data for research. This dissertation relies heavily on publicly available data published through the European Citizens' Initiative platform and combines this data with external socio-economic datasets. The work carried out by Lnenicka et al. (2023) supports the use of open government data as a reliable and valuable source of information for large scale analytical projects and further reinforces the importance of data integration within modern data analytics research (Lnenicka et al., 2023).

2.5 Direct democracy and equality: context is the key

Geißel, Krämling, and Paulus (2023) explore the relationship between direct democratic mechanisms and equality across European countries. By using large scale statistical models, the authors were able to investigate whether country level variables could explain outcomes associated with direct democratic processes. Their findings suggested that country level variables alone often fail to provide clear explanations for observed outcomes. The authors concluded that contextual and case specific factors frequently play a more significant role than broad socio-economic variables. As a result, they argue that statistical analysis should be complemented by an understanding of the social and political context in which participation occurs (Geißel et al., 2023).

This study is relevant to the dissertation as it demonstrates the limitations of relying solely on country level variables to explain participation behaviour. The findings support the possibility that variables such as GDP, education and population may only partially explain participation patterns observed within European Citizens' Initiatives. As a result, this study provides an important theoretical explanation for why predictive models based solely on socio-economic factors may produce relatively weak results despite substantial variation in participation across countries and initiatives (Geißel et al., 2023).

2.6 Research Gap

The reviewed research papers have demonstrated that significant research has been conducted on European Citizens' Initiatives, online petition participation, direct democracy and open government data. Existing studies have highlighted the importance of mobilisation strategies, campaign organisation and contextual factors in influencing participation outcomes. In addition, research has also shown that large scale datasets can be effectively analysed using quantitative methods to examine patterns of citizen engagement.

However there exists gaps that remain within the literature. First, much of the research with regards to European Citizens' Initiatives focus on democratic legitimacy, institutional effectiveness and campaign mobilisation rather than quantitative analysis of participation behaviour. Secondly, while online petition studies have been conducted with statistical methods, they often are limited to national platforms and do not consider participation across multiple countries. Finally, limited research has been conducted that combines European Citizens' Initiative signature data with external socio-economic factors in order to investigate whether or not demographic and economic variables influence participation rates.

This dissertation seeks to address these gaps by employing web scraping techniques to collect ECI signature data, and integrate that data with external socio-economic variables, and then applying exploratory and predictive analytical methods. Through this approach, this dissertation aims to contribute to new research examining political participation through the lens of data analytics.

3 Exploration of Data and Methods

3.1 Research Methodology

This project adapts from the Cross Industry Standard Process for Data Mining (CRISP-DM) methodology. CRISP-DM provides a structured framework for data analytics research through a series of iterative stages such as business understanding, data understanding, data preparation, modelling, evaluation and deployment (Data Science PM, 2024).

This methodology was chosen because of its widespread use within data analytics projects and its suitability for integrating multiple data sources (Data Science PM, 2024). This project currently focuses on the earlier stages of CRISP-DM being data collection, preparation, exploration and initial modelling. More advanced modelling and evaluation is planned for the final dissertation.

The overall workflow adopted for this project is illustrated in (Figure 1).

3.2 Data Collection

Two separate data sources were used for this project:

European Citizens' Initiative Data

The ECI signature datasets were collected from the European Citizens' Initiative website.

A web scraping script was developed using Python and the Playwright framework (Medium, 2025). The scraper manually opens a chromium session on the selected page and extracts the table data from the initiative web page, extracting the country level signature data and exporting it as a CSV file. This process was repeated for 7 Initiatives for this stage of the project.

The following Initiatives were selected:

- *Ban on Conversion Practices in the European Union* (2024)
- *European Citizens' Initiative in Defence of Agriculture and Rural Economy in Europe* (2024)
- *My Voice, My Choice: For Safe and Accessible Abortion* (2024)
- *Air-Quotas* (2024)
- *Stop Destroying Videogames* (2024)
- *Stop Cruelty Stop Slaughter* (2024)
- *Stop Fake Food: Origin on Label* (2024)

The extracted variables from each Initiative:

- Country Name
- Number of Signatures
- Country Minimum Threshold
- Percentage of Threshold Reached

Each Initiative was exported into a separate CSV file.

External Socio-Economic Dataset

A secondary dataset was obtained from the Kaggle Worldstat (Elgiriye withana, 2023) dataset, this dataset contains country-level statistics and demographic, economic and social variables for countries worldwide.

The following variables from this dataset were selected:

- Country Name
- Population
- GDP
- Unemployment Rate
- Life Expectancy
- Population Density
- Gross Tertiary Education Enrollment
- Urban Population
- Labor Participation

Only European Union countries appearing within the ECI datasets were retained from this Dataset for analysis.

3.3 Ethical Considerations

This project uses publicly available data sourced from official European Union websites and publicly available statistical datasets.

No personally identifiable information was collected during the web scraping process. All ECI data is aggregated at the country level and therefore does not contain any individual personal information.

The scraping process was conducted with responsible data collection principles by collecting only publicly available information and also keeping data scraping to a minimum to avoid overburdening the websites' resources. All was stored securely and used exclusively for analysis within the scope of the dissertation.

A DkIT ethical Approval form was completed and sent before the conducting any research, to ensure that the project follows DkIT ethical research guidelines.

This project's research follows closely the ACM code of ethics to ensure research remains transparent and honest (Association for Computing Machinery, 2018). As no personal data was collected, the project ensures minimal privacy risk.

3.4 Initial Exploratory Data Analysis

Initial EDA was conducted to explore each dataset separately, to identify key variables, find missing values, duplicate rows, dataset dimensions, column names etc. to determine future data cleaning and merging as well as feature extraction. It was found that most numerical variables contained string data and would need to be cleaned appropriately and converted to numerical. Some country names were inconsistent to those in the Worldstat dataset. Many numerical variables had missing values in some of the rows of the Worldstat dataset. Each dataset will need to be cleaned first before merging, with names being standardised and then using country name to merge the datasets with the Worldstat data.

3.5 Data Preparation

The following data preparation steps were undertaken to create a unified dataset ready for analysis:

ECI Datasets:

1. ECI datasets were imported into a single dataframe
2. Removal of non country specific records
3. Removal of symbols and commas

4. Conversion of objects to numericals
5. Standardisation of column names
6. Standardisation of Country names to match the Worldstat dataset

Worldstat Dataset:

1. Imported into a dataframe with only the selected variables
2. Checking and handling of missing values
3. Checking and handling of Duplicates
4. Checking and handling of nonsense values, converted into NaN
5. Another check on missing values after converting nonsense values to NaN
6. Outlier detection using IQR method
7. Removal of percentage symbols, currency symbols and commas
8. Conversion of objects to numericals
9. Standardisation of column names

None of the EU countries had missing values for the selected variables making it safe to merge.

Unified Dataset:

1. Merged ECI dataset with Worldstat Dataset using Country name
2. Double checking for missing values
3. Double checking for duplicate records
4. Previewing of unified dataset

Feature Engineering:

The following additional variables were created:

- Signatures Per Capita = Signatures / Population
- GDP Per Capita = GDP / Population
- Threshold Ratio = Signatures / Threshold
- Threshold Per Capita = Threshold / Population

The unified dataset was then previewed and checked one last time before being exported as a CSV file.

3.6 Post Cleaning Exploratory Data Analysis

Further EDA was conducted to identify patterns, trends and relationships with the data.

Univariate Analysis

This analysis revealed:

- Population Density was heavily right skewed (Figure 5)
- GDP per Capita was heavily right skewed (Figure 3)
- Raw signature counts showed significant variation between countries (Figure 7)
- Signatures per capita also showed substantial variability between initiatives (Figure 8)

Correlation Analysis

Pearson, Spearman and Kendall's Tau analyses were performed (Figures 13, 14, 15):

- Population vs Signatures = 0.45
- GDP vs Signatures = 0.43
- GDP per Capita vs Signatures = 0.04
- Life Expectancy vs Signatures = 0.20
- Population Density vs Signatures = 0.01
- Unemployment Rate vs Signatures = 0.14

This revealed multicollinearity:

- Population vs Threshold = 1.00
- Threshold Percentage vs Threshold Ratio = 1.00

These variables were identified as candidates for removal in later stages of the project

Visualisations

Scatterplots were created:

- Signatures per Capita vs Population Density (Figure 12)
- Signatures per Capita vs GDP per Capita (Figure 9)
- Signatures per Capita vs Unemployment (Figure 10)
- Signatures per Capita vs Education (Figure 11)

Many of these relationships appeared to be weak or non-linear.

3.7 Initial Data Modelling

Initial Modelling was conducted to investigate the relationships further and to answer the Hypotheses.

Analysis of Variance (ANOVA)

A one-way ANOVA was conducted to test for variance in signatures across different petitions:

The test produced a p-value < 0.05

This suggests that participation levels differ greatly across initiatives, suggesting that petition-specific factors may play an important role in determining participation, rejecting the Null Hypothesis.

Linear Regression Model 1

The first model examined whether Population and GDP could explain some of the variance in total signature counts.

$$\text{Signatures} = \text{Population} + \text{GDP}$$

This model produced an R squared value of 0.256 meaning that it explained 25.6% of the variance in Signatures.

This suggests that larger and wealthier countries generally have more signatures, rejecting the Null Hypothesis.

Linear Regression Model 2

This second model examined participation rates after controlling for population size.

Signatures Per Capita GDP Per Capita + Unemployment + Life Expectancy + Population Density + Tertiary Enrollment

This model produced an R-Squared value of -0.045 meaning it failed to explain variance in Signatures per Capita.

This suggests that socio-economic factors alone are insufficient for explaining variation in participation rates. Failing to reject the Null Hypothesis.

3.8 Initial Evaluation

The modelling results indicated that petition specific factors may be more important than country level socio-economic factors when explaining participation rates.

This conclusion is consistent with the findings reported by Weisskircher (2019) and Giebler et al. (2025) who both emphasised the importance of campaign context and issue specific factors.

3.9 Artefact Design

The artefact developed during this stage of the project is the Python based data collection and analysis pipeline.

Figure 1 presents a diagram of the architecture of the artefact.

This modular design allows for additional initiatives to be incorporated easily as the project evolves.

4 Proposed Future Analysis

The next stage of the project will focus on expanding both the dataset and the models used. More ECIs will be scraped from the ECI platform to increase the number of observations and improve the robustness of the analysis. Further data preparation and feature engineering will be undertaken, including the incorporating new variables such as media literacy, voter turnout, trust in government etc, as well as possibly explore petition specific factors.

New models are planned, including Principal Component Analysis (PCA) for dimensionality reduction and clustering methods to identify groups of countries with similar participation patterns. Additional regression models and machine learning models will also be incorporated and compared.

This final dissertation will hopefully achieve a more comprehensive understanding of participation in European Citizens' Initiatives.

Bibliography

- Air-quotas*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000006_en.
- APXML. (n.d.). *Using Joblib persistence*. <https://apxml.com/courses/basics-ml-deployment/chapter-2-preparing-model-deployment/using-joblib-persistence>. APXML.
- Association for Computing Machinery. (2018). *Acm code of ethics and professional conduct*. Retrieved from <https://www.acm.org/code-of-ethics> (Accessed: 07 April 2026)
- Ban on conversion practices in the European Union*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000001_en.
- Data Science PM. (2024). *What is crisp dm? data science project management*. <https://www.datascience-pm.com/crisp-dm-2/>.
- Elgiriye withana, N. (2023). *Global country information dataset 2023*. Data set. Kaggle. Retrieved from <https://doi.org/10.34740/KAGGLE/DSV/6101670> doi: 10.34740/KAGGLE/DSV/6101670
- Europa.eu. (2024). *Thresholds*. European Citizens' Initiative. Retrieved from https://citizens-initiative.europa.eu/thresholds_en
- European Commission. (n.d.). *History of the european citizens' initiative*. https://citizens-initiative.europa.eu/how-it-works/history_en.
- European citizens' initiative in defence of agriculture and rural economy in Europe*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000002_en.
- Geißel, B., Krämling, A., & Paulus, L. (2023). *Direct democracy and equality: context is the key*. <https://doi.org/10.1057/s41269-023-00316-4>.
- Giebler, H., Giesecke, J., Humphreys, M., Hutter, S., & Klüver, H. (2025). *Mobilizing eu-rope's citizens to take action on migration and climate change: behavioral evidence from 27 eu member states*. <https://doi.org/10.1080/13501763.2025.2512032>.

- Hale, S. A., Margetts, H., & Yasseri, T. (2013). *Petition growth and success rates on the uk no. 10 downing street website*. <https://doi.org/10.1145/2464464.2464518>.
- Lnenicka, M., Nikiforova, A., Luterek, M., Milic, P., Rudmark, D., Neumaier, S., ... Pedro, M. (2023). *Identifying patterns and recommendations of and for sustainable open data initiatives: a benchmarking-driven analysis of open government data initiatives among european countries*. <https://arxiv.org/abs/2312.00551>.
- Medium. (2025). *How to scrape websites with Playwright and Python*. <https://medium.com/@hasdata/how-to-scrape-websites-with-playwright-and-python-49a015fd00aa>.
- My voice, my choice: For safe and accessible abortion*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000004_en.
- Stop cruelty stop slaughter*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000008_en.
- Stop destroying videogames*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000007_en.
- Stop fake food: Origin on label*. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000009_en.
- Weisskircher, M. (2019). *The european citizens' initiative: Mobilization strategies and consequences*. <https://doi.org/10.1177/0032321719859792>.

A Appendices

A.1 Project Workflow Chart

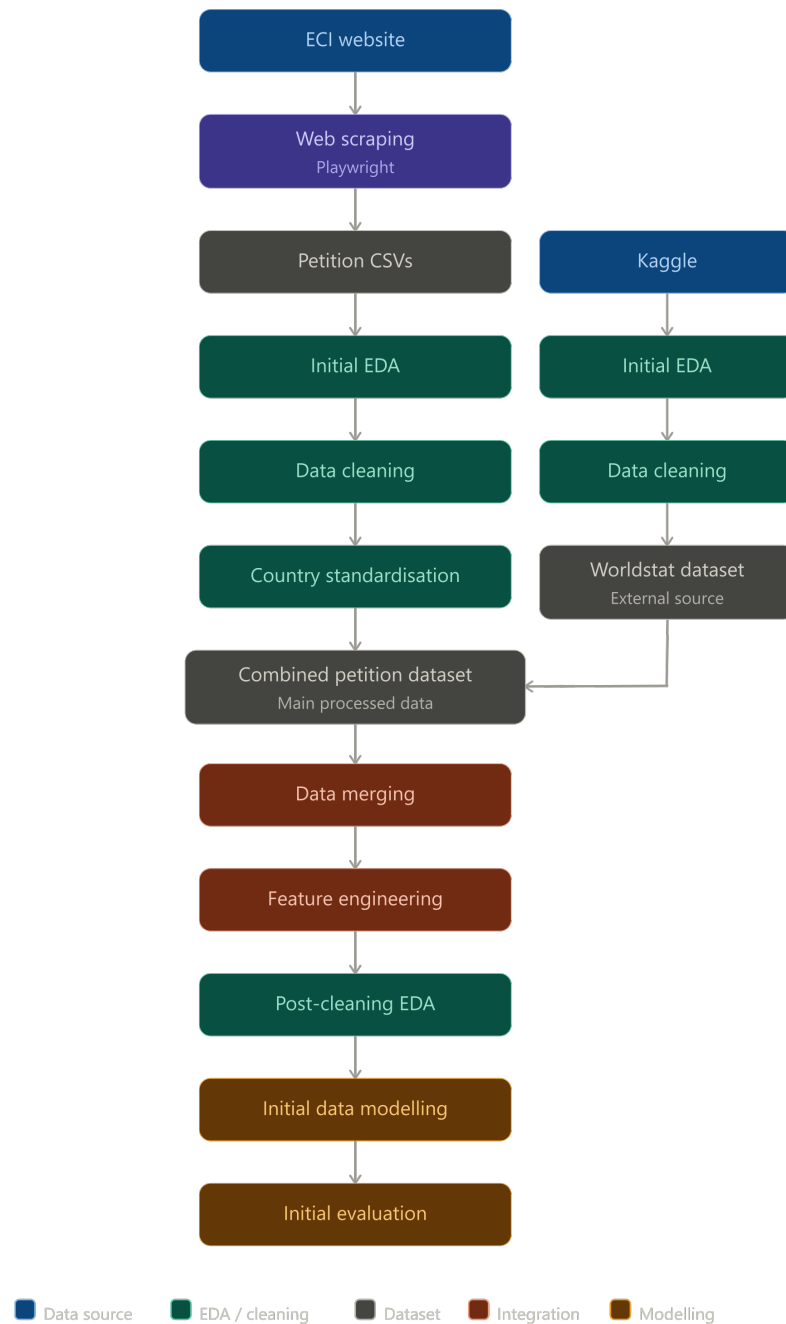


Figure 1: Project Workflow Chart

A.2 Histogram of Signatures Per Capita

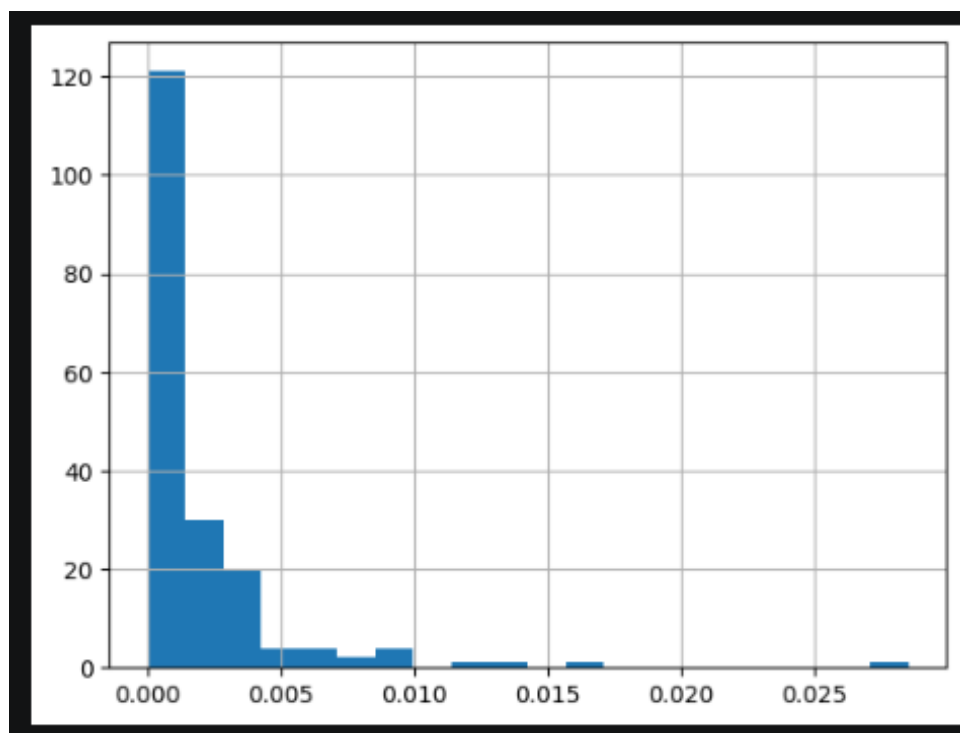


Figure 2: Signatures per Capita

A.3 Histogram of GDP per Capita

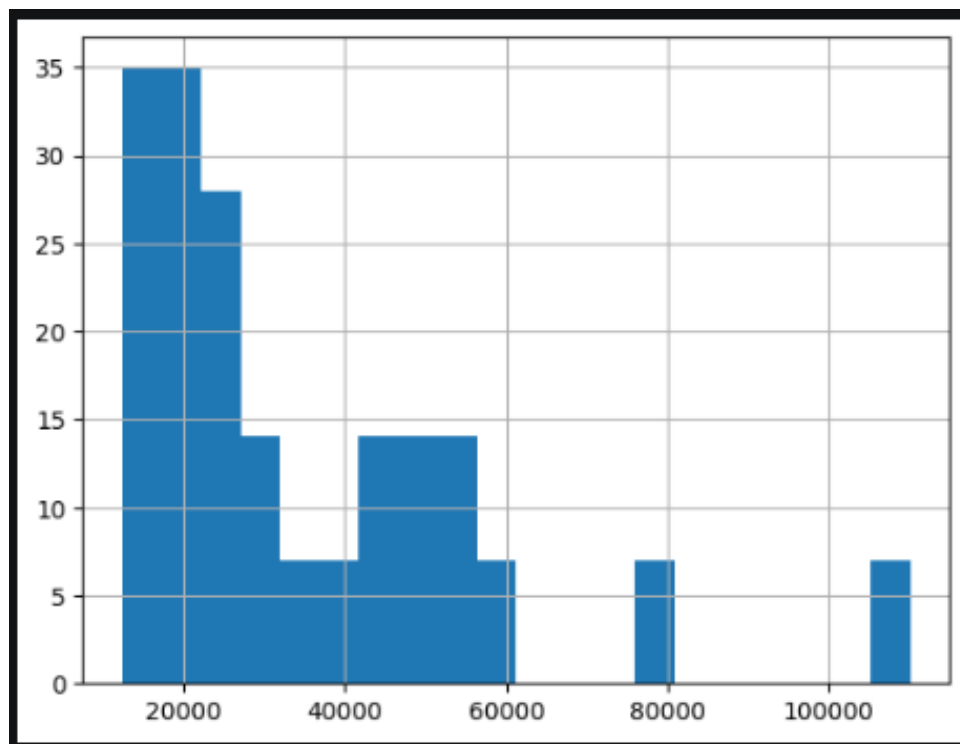


Figure 3: GDP per Capita

A.4 Histogram of Unemployment Rate

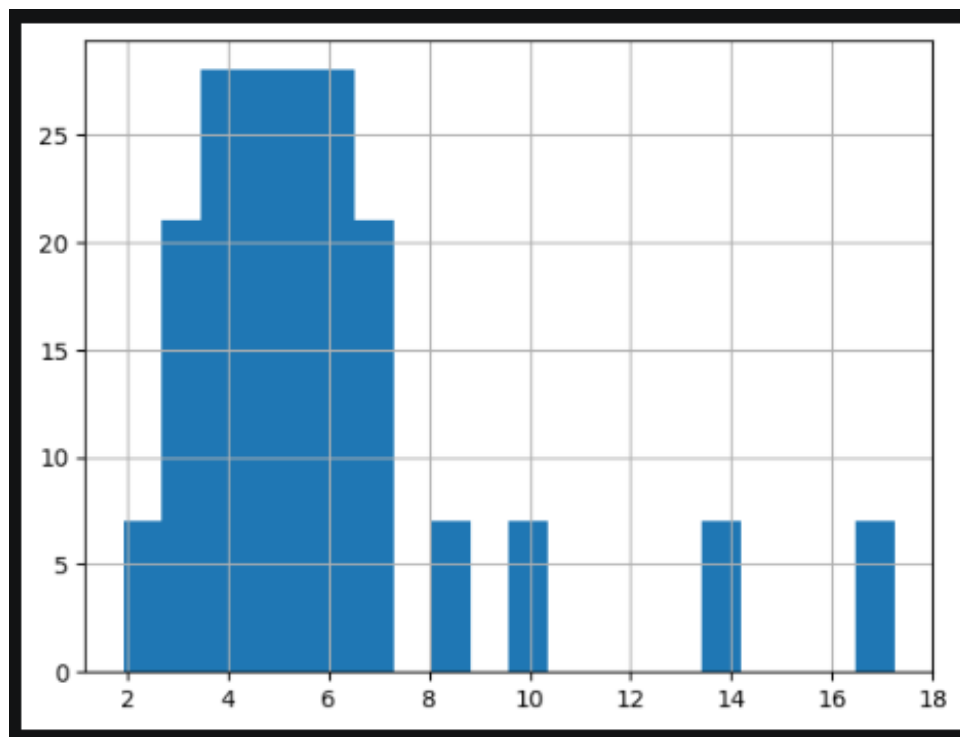


Figure 4: Unemployment Rate

A.5 Histogram of Population Density

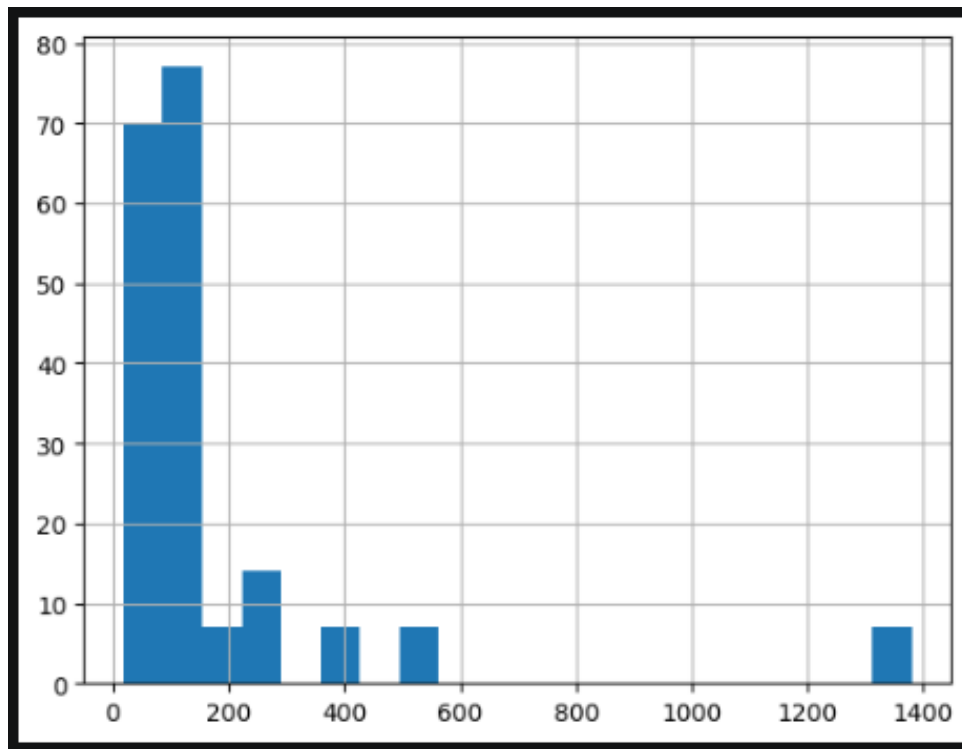


Figure 5: Population Density

A.6 Boxplot of GDP Per Capita

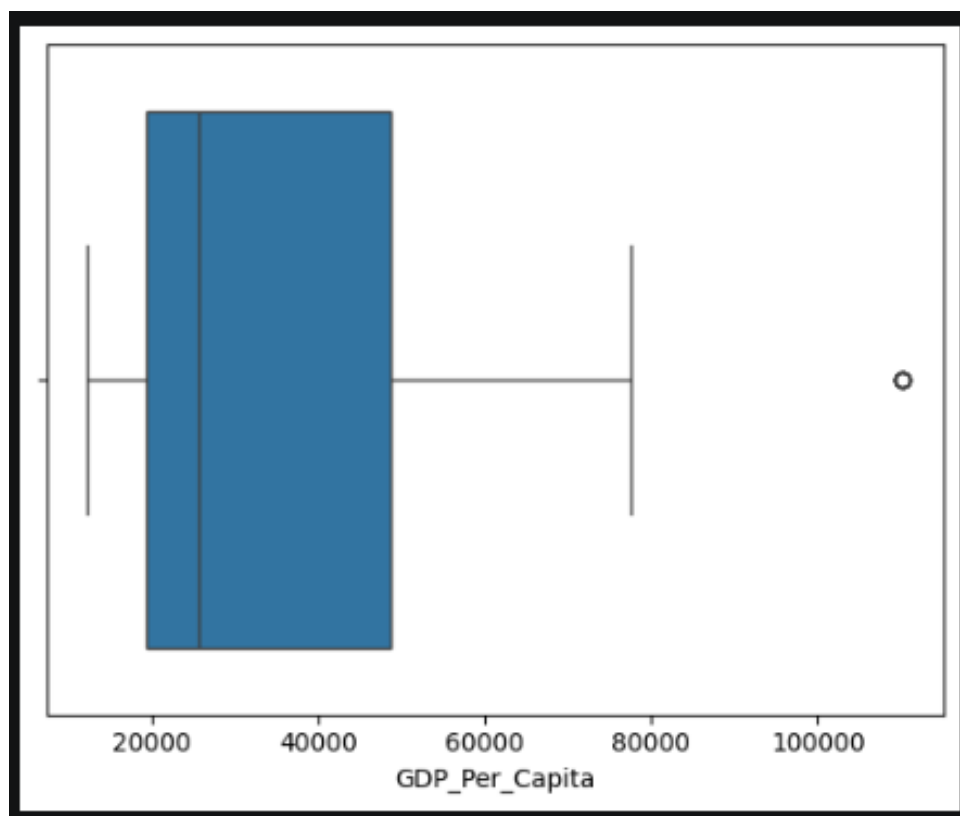


Figure 6: Boxplot of GDP Per Capita

A.7 Countries with highest Signatures per Capita

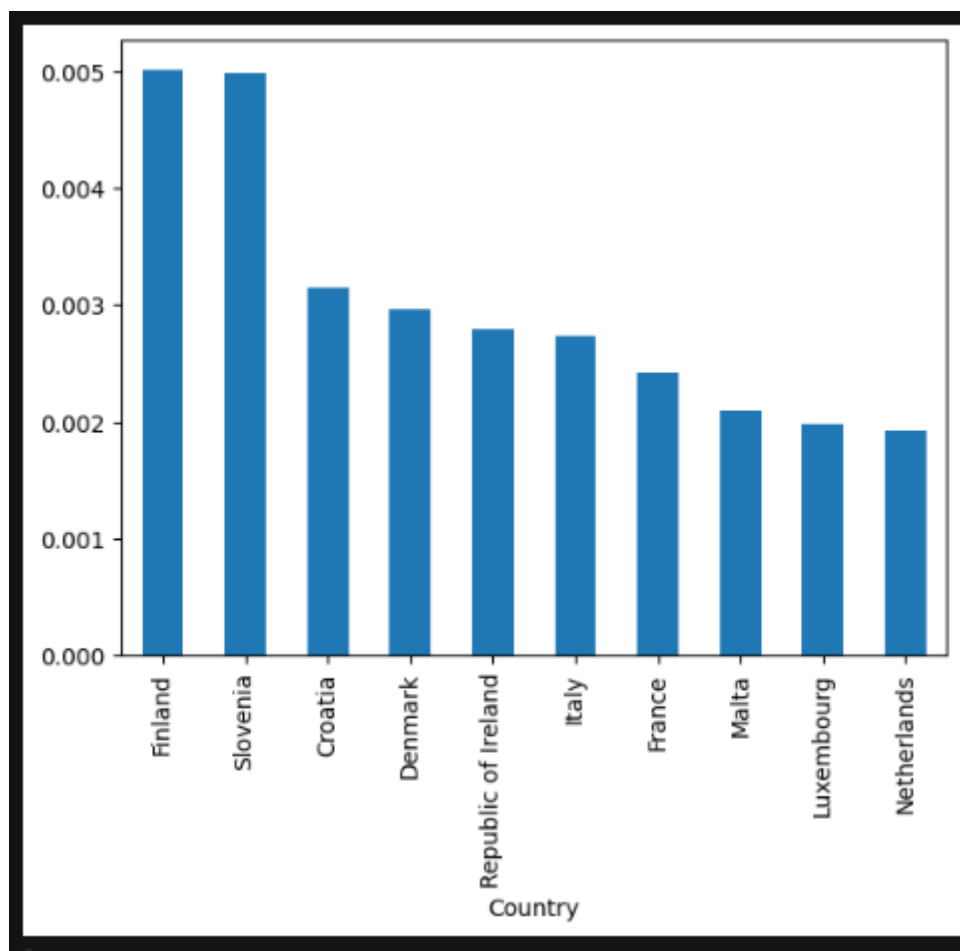


Figure 7: Countries with highest Signatures per Capita

A.8 Signatures per Capita vs Petition

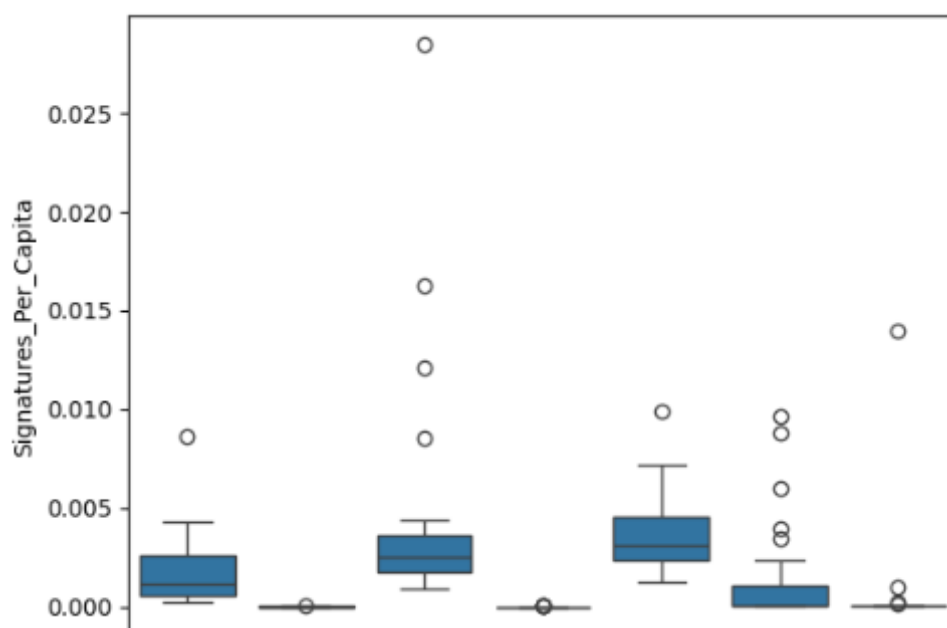


Figure 8: Signatures per Capita vs Petition

A.9 Signatures per Capita vs GDP per Capita

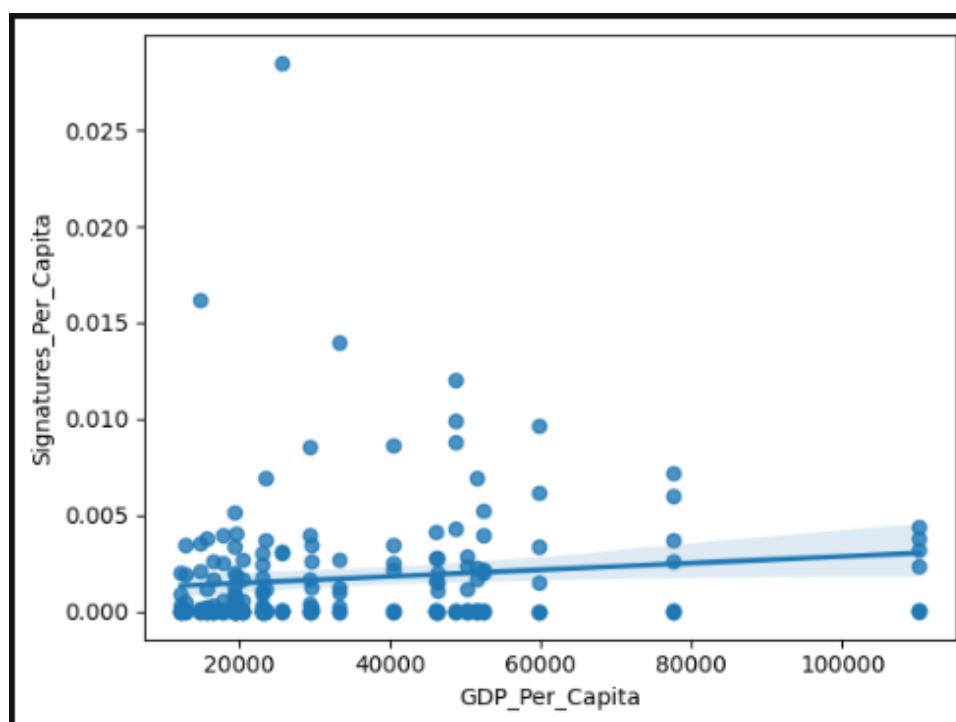


Figure 9: Signatures per Capita vs GDP per Capita

A.10 Signatures per Capita vs Unemployment Rate

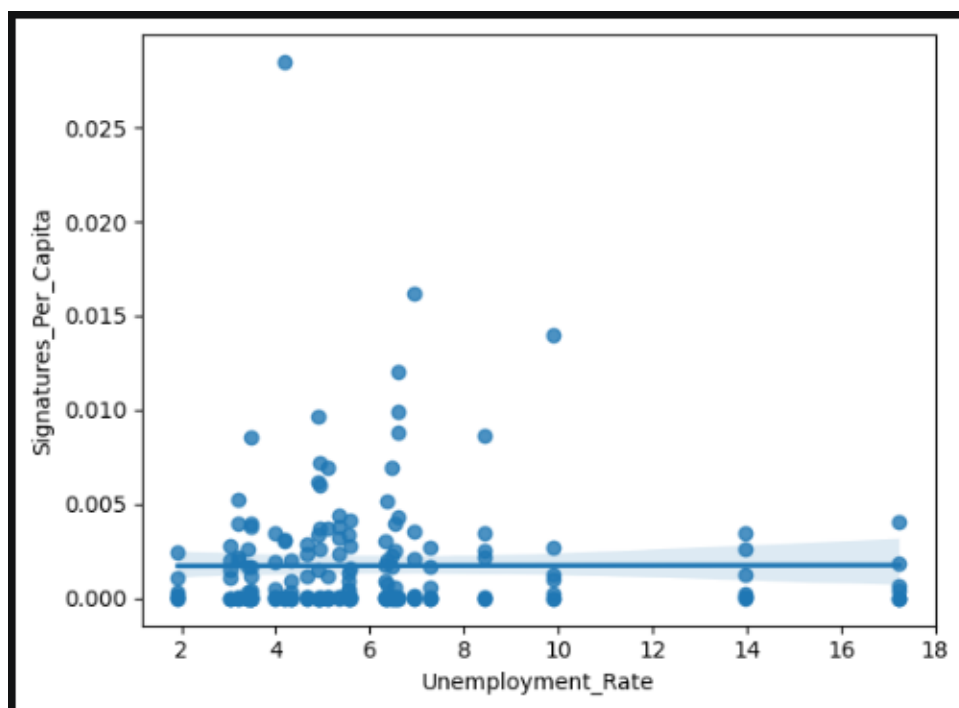


Figure 10: Signatures per Capita vs Unemployment Rate

A.11 Signatures per Capita vs Tertiary Enrollment

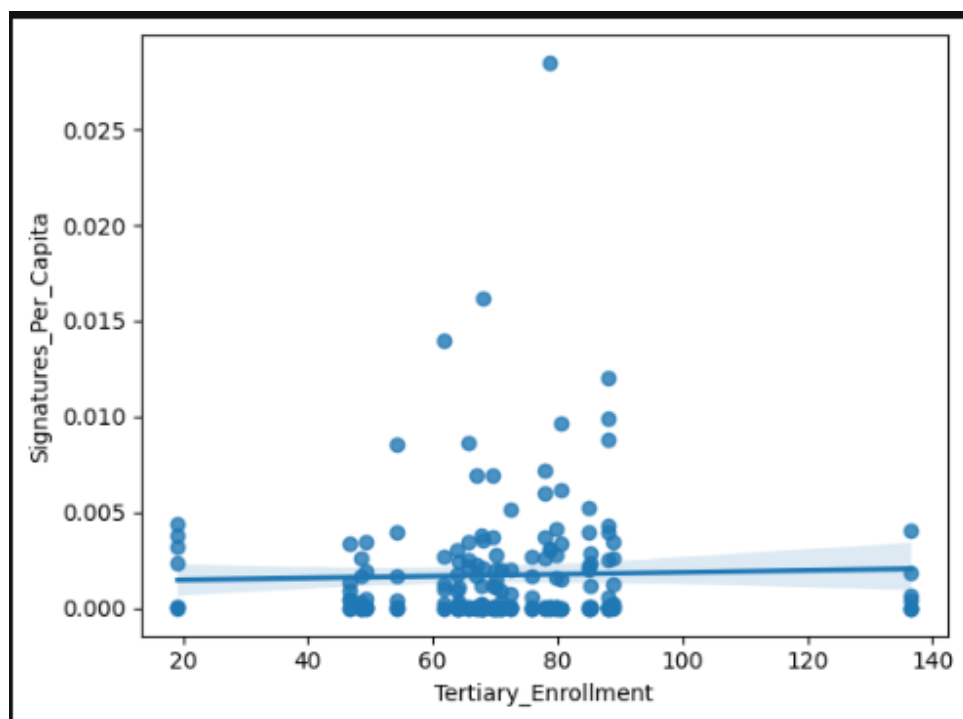


Figure 11: Signatures per Capita vs Tertiary Enrollment

A.12 Signatures per Capita vs Population Density

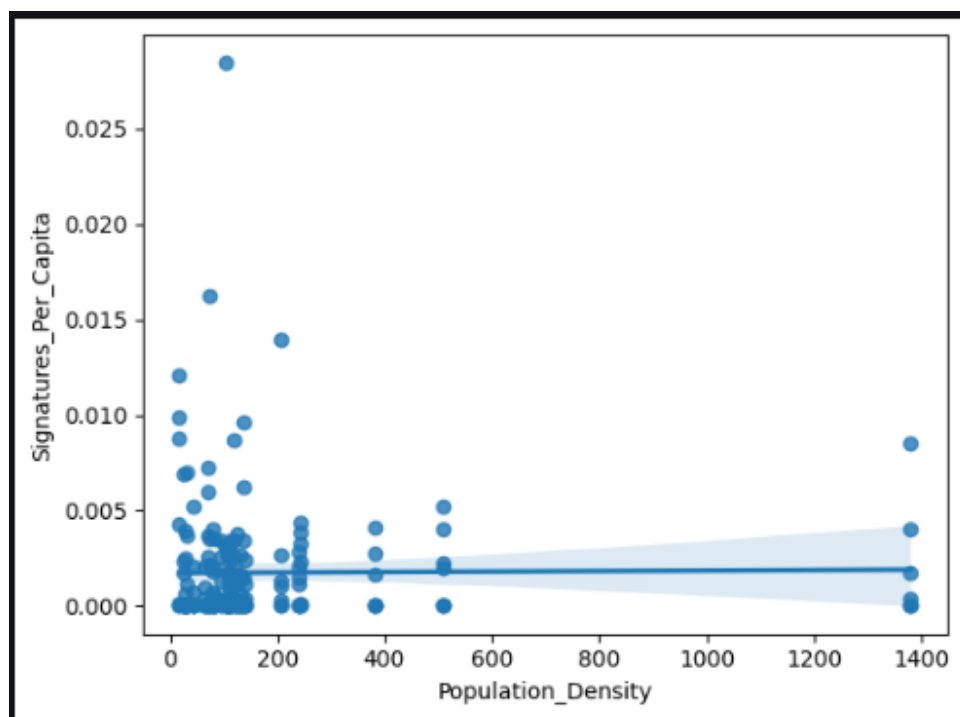


Figure 12: Signatures per Capita vs Population Density

A.13 Pearson Heatmap

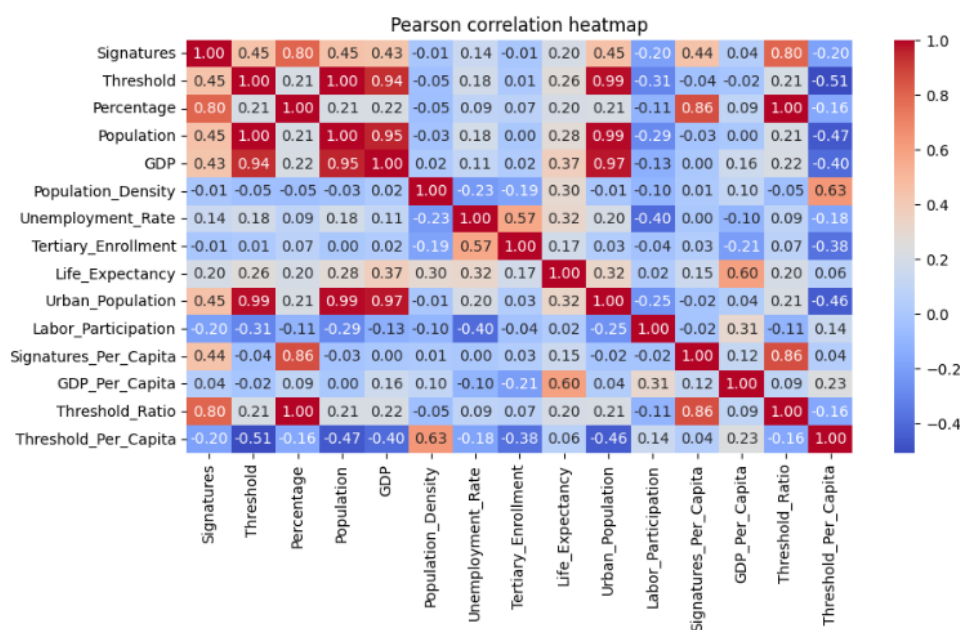


Figure 13: Pearson Heatmap

A.14 Spearman Heatmap

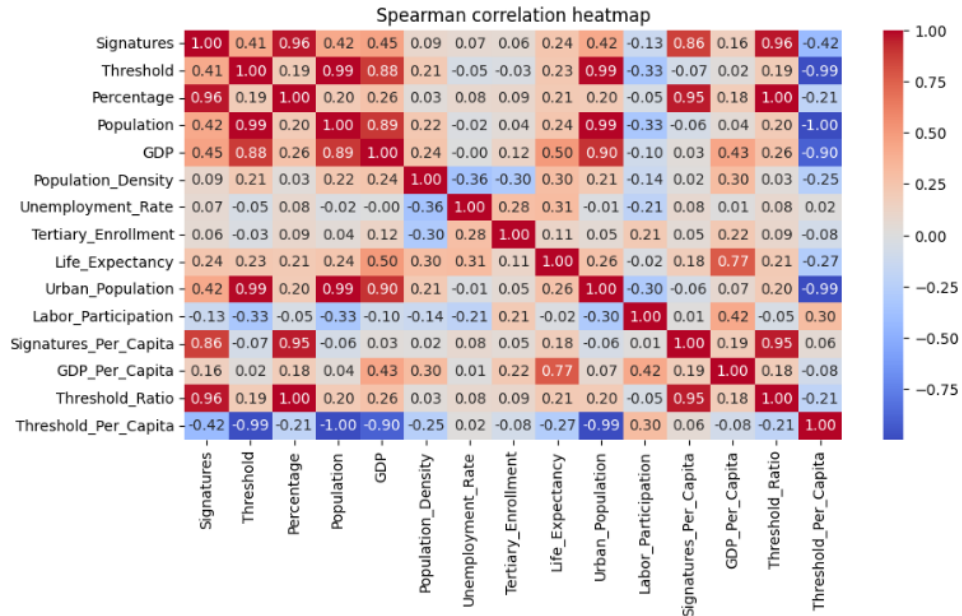


Figure 14: Spearman Heatmap

A.15 Kendall's Tau Heatmap

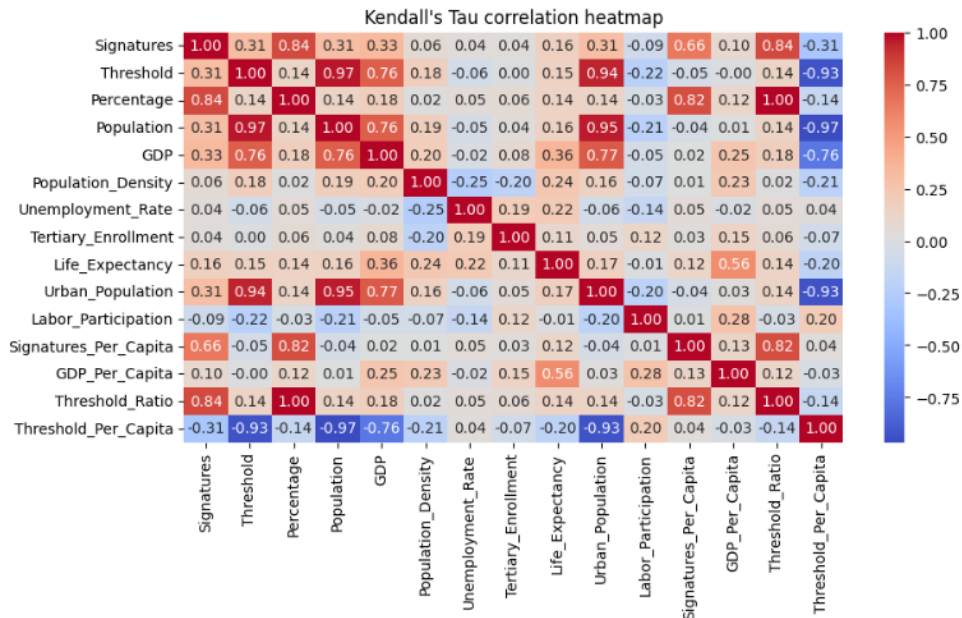


Figure 15: Kendall's Tau Heatmap

A.16 Github Repository

<https://github.com/Matthew-Riddell/MSc-Dissertation>

A.17 Mahara Portfolio

<https://mahara.dkit.ie/view/view.php?t=f816cb6da6dd833aaa2f>